

Knowledge Representation

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Outline

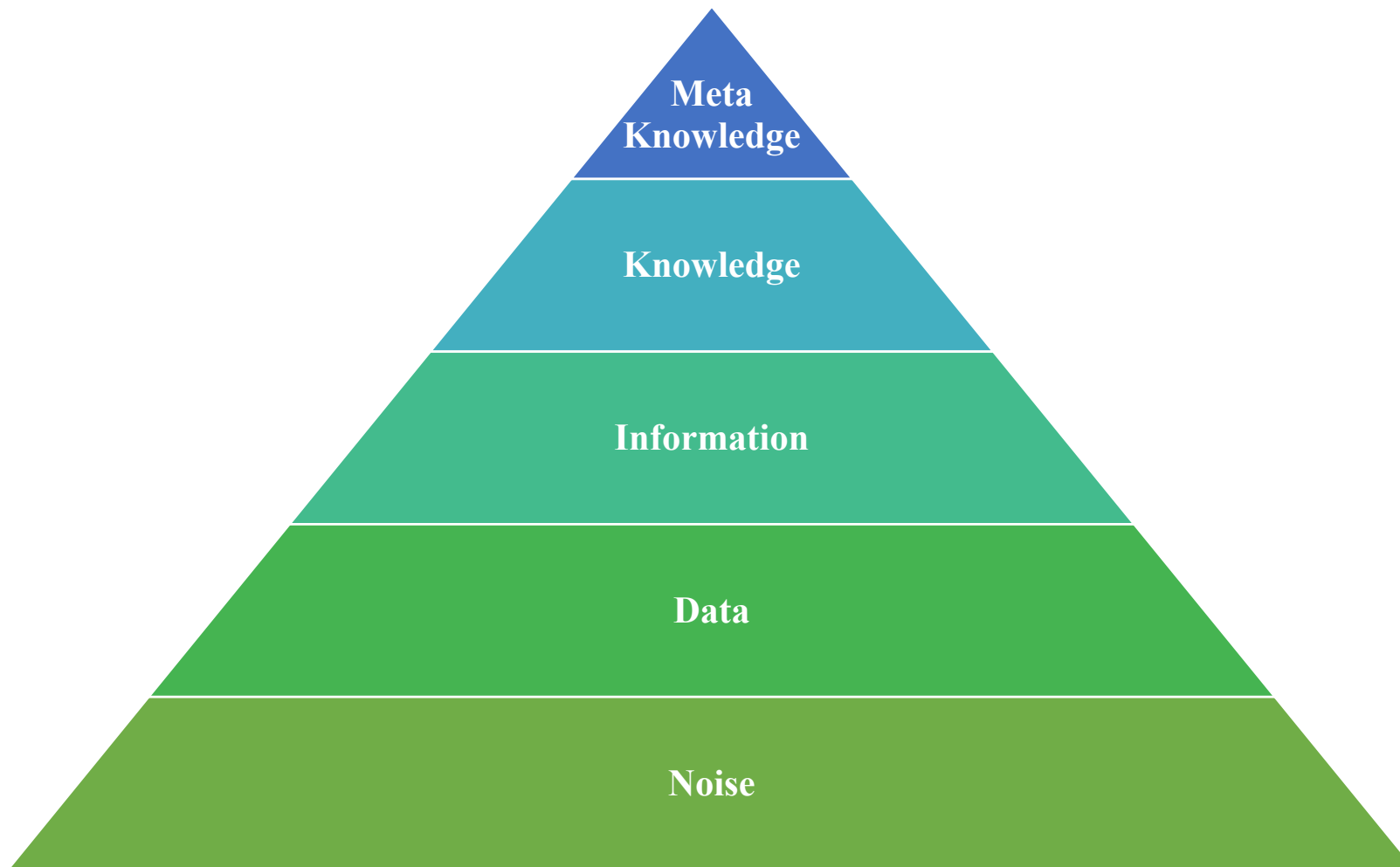
- What is knowledge?
- How does knowledge involve in AI problems?
- Knowledge representation methods
 - Propositional logic
 - First-order logic
 - Semantic networks

What is knowledge?

- **Symbolism:** Knowledge is the foundation of human intelligence
- Human intelligent comes from a process of acquiring and applying knowledge
- Solutions to AI problems are based on knowledge. The **acquisition**, **representation** and **application** of knowledge for reasoning are the three main problems of AI research.

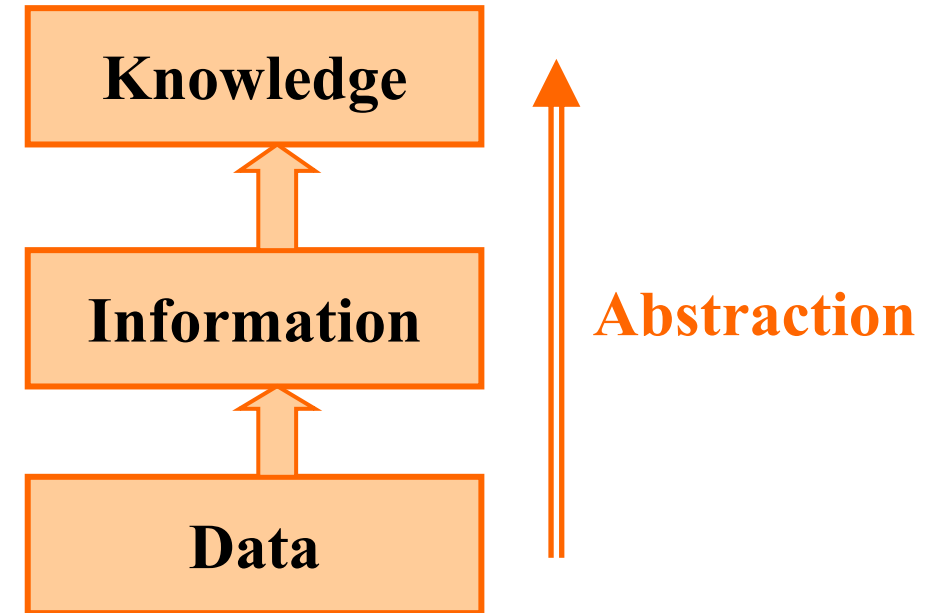


Knowledge Pyramid

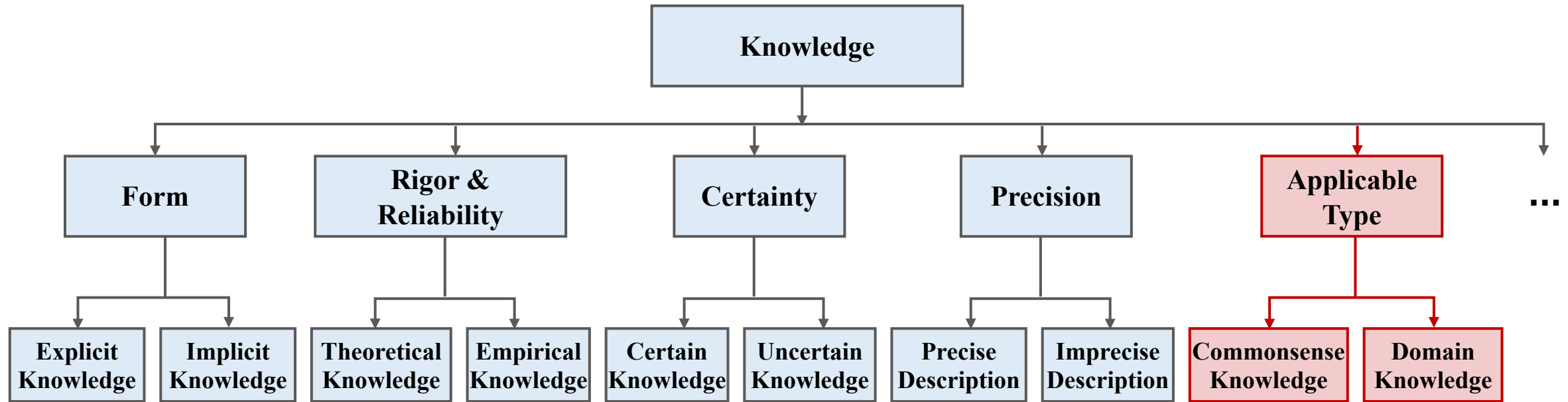


Data, Information and Knowledge

- **Data**: symbols that record information
 - E.g., 71 79 76 68 32 49 54 53 43
- **Information**: interpretation of data within certain context.
 - E.g., 71 79 76 68 32 49 54 53 43 → GOLD 165+
- **Knowledge**: a structure that covers relevant information
 - E.g., “GOLD 165+” Probably mean the gold price is above 165 USD per gram



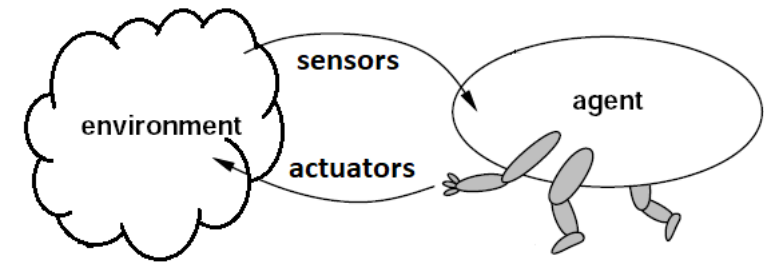
Categorisation of Knowledge



Knowledge in AI Problems

Intelligent Agents

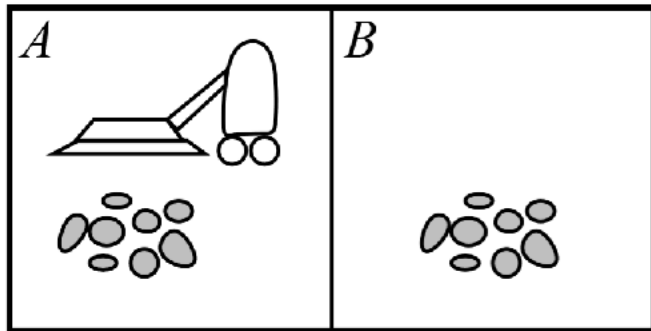
- A computational **agent** is an entity capable of
 - Perceiving the environment
 - Deliberating
 - Performing actions
- **Artificial Intelligence** can be viewed as the study of the synthesis and analysis of computational agents that act intelligently
- An **agent task** is defined by:
 - **Environments**: The states of the task
 - **Sensors**: Representations of information about the environment states that are input to the agent system
 - **Actuators**: Actions that may modify the environment states
 - **Performance measure**: Evaluation function of any given sequence of environment states



Intelligent Agents

- **Percept** is the agent's perceptual input at any given instant
- An agent's **percept sequence** is the complete history of everything that the agent has perceived.
- An **agent program** maps a given percept sequence to an action. This function describes the agent's behaviours.

Example [vacuum world].

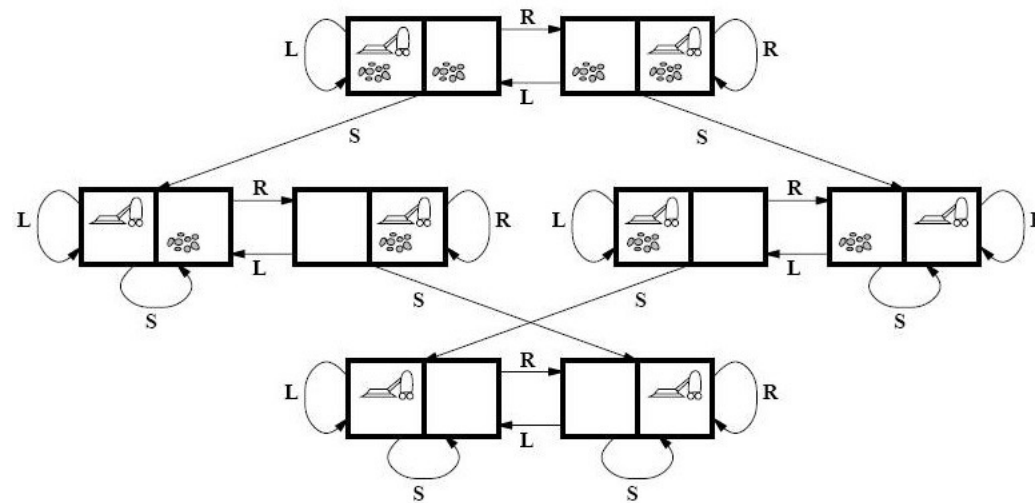


Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$

Agents with a World Model

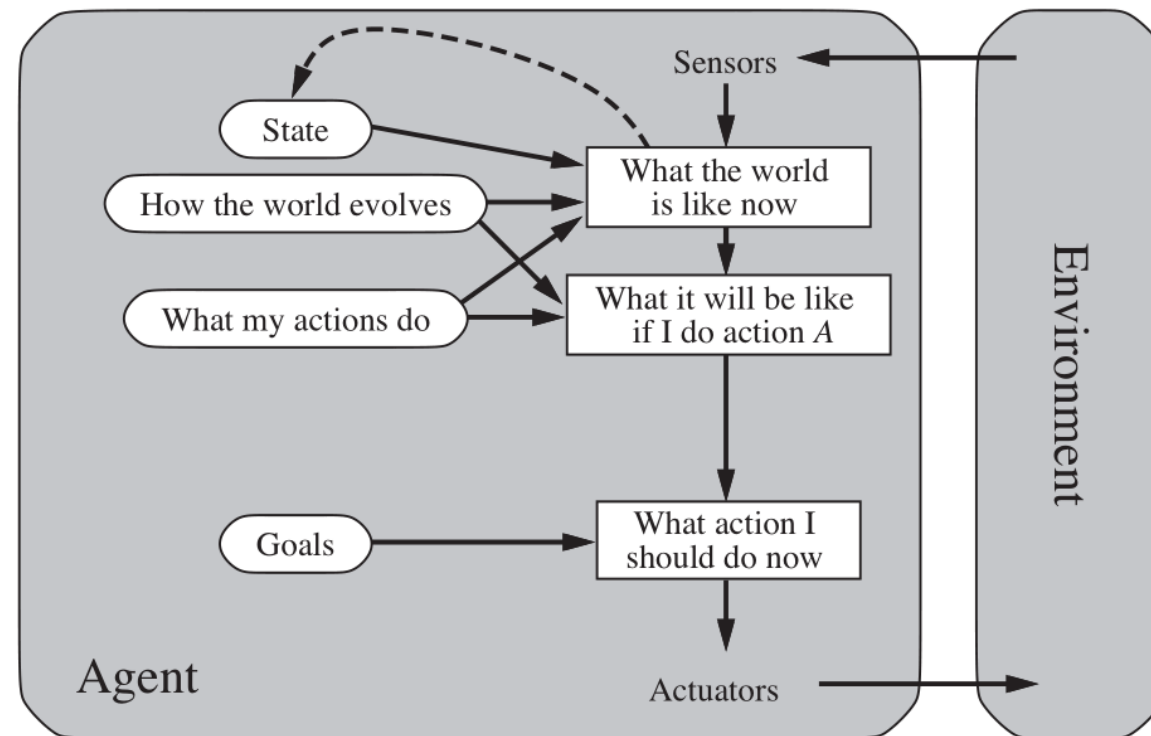
- **World model**: Encodes information about
 - How the world evolves independently of the agent;
 - How the agent's actions affect the world.
- A **model-based agent program** uses a world model to determine the agent's actions. In particular
 - The agent maintains an **internal state** in the world model
 - The internal state is updated by a **state-transition function** based on the agent's percept history
 - An agent's behaviours may be directed by **goals** or **utilities**.

Example. The following is a state-based world model of the vacuum world



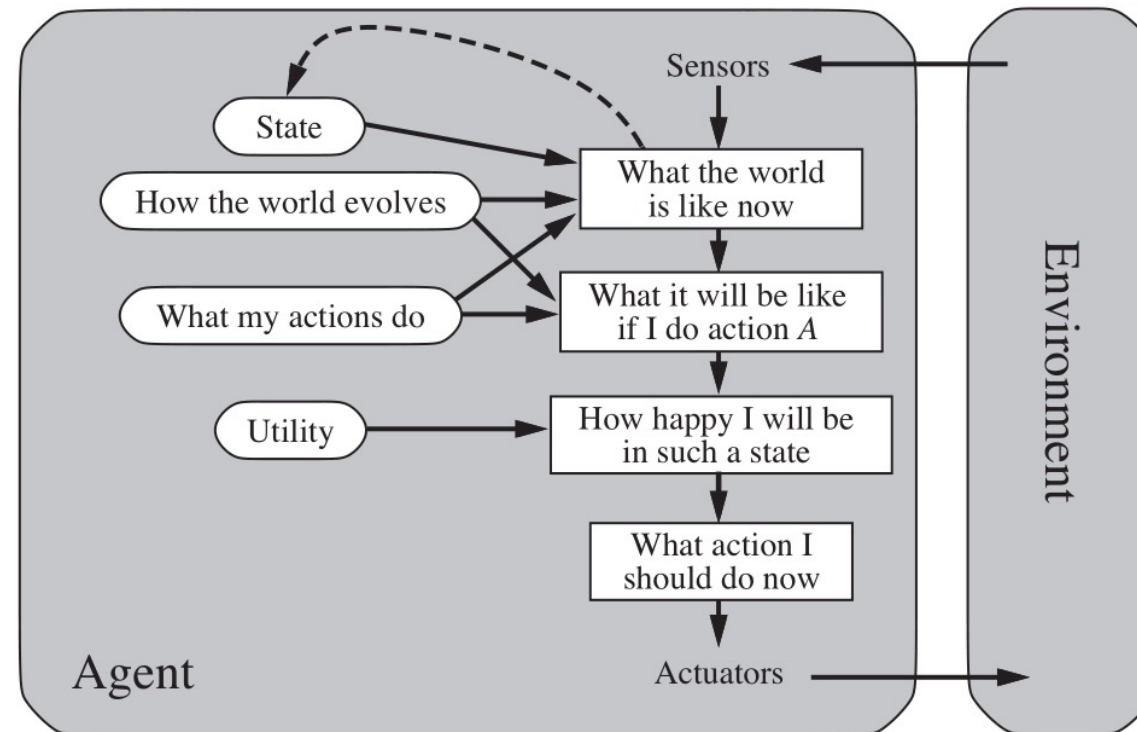
Goal-based Agents

- A **goal-based** agent program maintains explicit information about the situations that are desired by the agent.
E.g., In the vacuum world, the goal could be to achieve a state where all locations are clean



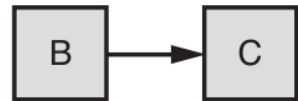
Utility-based Agents

- A **utility-based** agent program uses a utility function to measure how desirable a state is, and actions are derived to maximise utility.
E.g., In the vacuum world, a utility function may be the number of dirty locations cleaned.

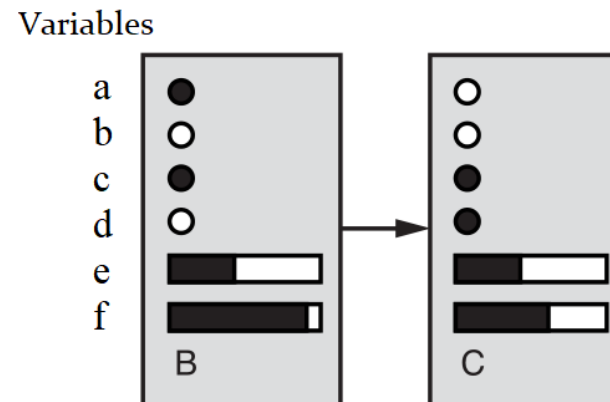


Representations of world models

- **Atomic representation:** Explicitly represent each state of the world and their transitions.
 - States are atomic
 - Any action leads to a state transition
- **Factored representation:** A state of the world is defined by a collection of features (or variables) and their values.
 - States are variable assignments
 - Any location leads to some potential changes to the variable values



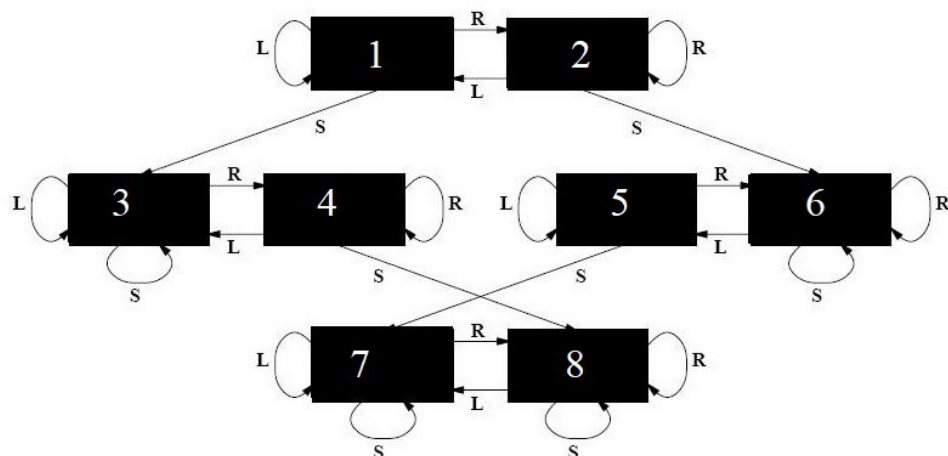
(a) Atomic



(b) Factored

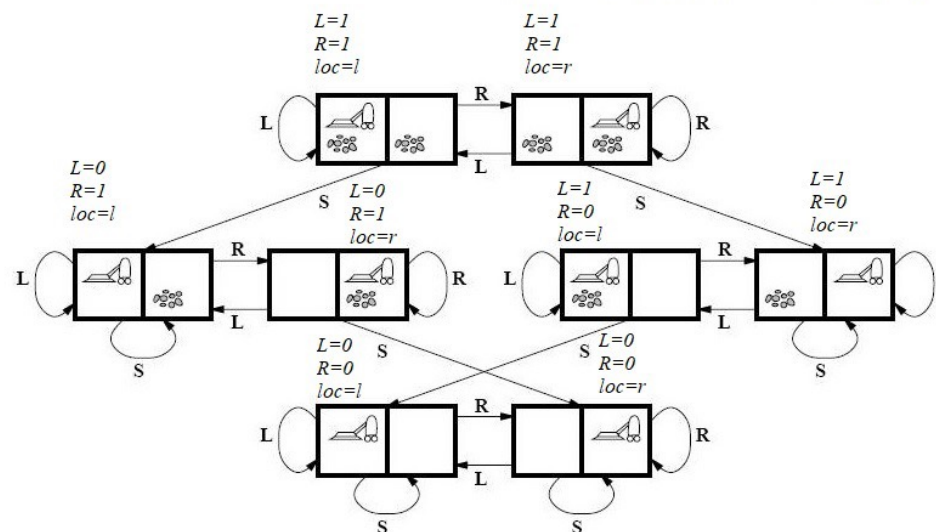
Example. The vacuum world may have the following atomic and factored representations:

States: $\{1,2,3,4,5,6,7,8\}$



Atomic Representation

Variables: L, R, loc **Domains:** $\text{Dom}(L)=\text{Dom}(R)=\{0,1\}$ $\text{Dom}(loc)=\{l,r\}$



Factored Representation

Structured Representation

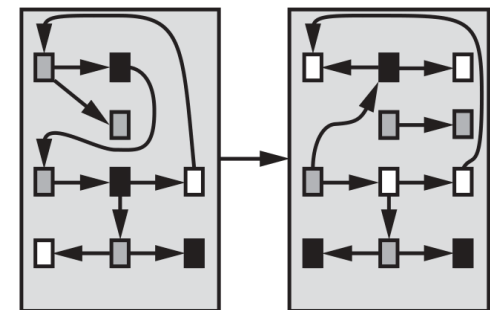
It is often insufficient to keep the variable values.

- An agent may only have partial knowledge about the state
- The variables may correlate with each other through constraints, thus the potential for reasoning. E.g., Being a fish & Living under water.

Structured representation

A **structured representation** of a world model describes the world using variables, while capturing the following:

- **Knowledge**: Constraints on variables
- **Reasoning**: Inference engines to derive knowledge from percepts

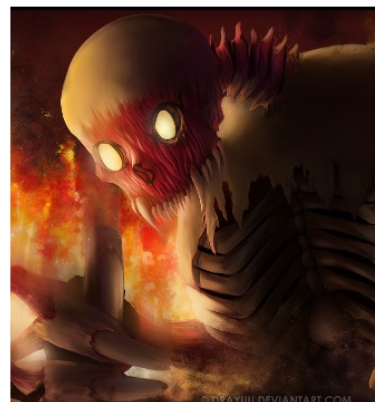
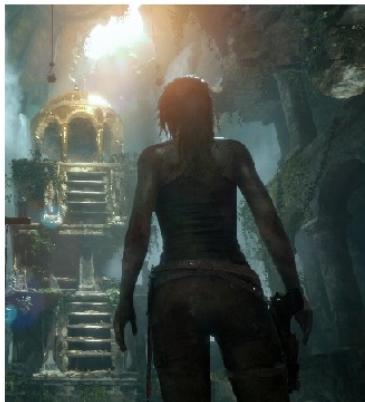


Example. [Wumpus world] An agent enters a tomb.

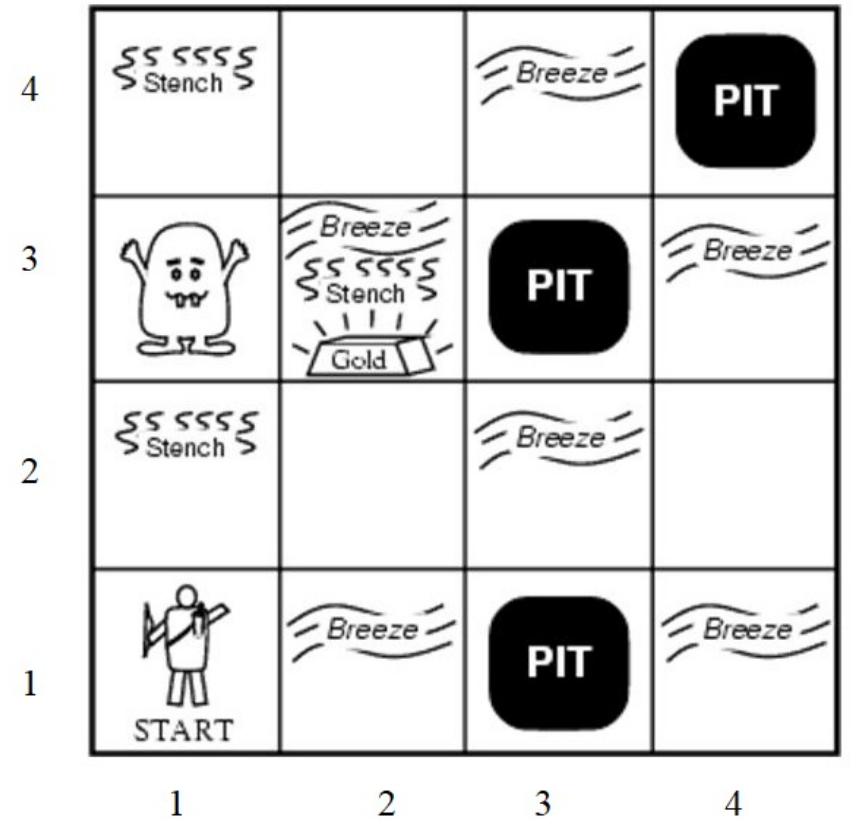
- A **wumpus** is a monster that eats the agent if the agent walks in the same cell.
- A **pit** is a trap that kills the agent if the agent steps on it.
- A **gold** is a treasure that the agent wants to get.

The agent can **perceive** information about her current location:

- The wumpus is stink, and the agent can sense **stench** when walking beside it.
- Wind blows out of the pit, and the agent can sense **breeze** when walking beside a pit.
- The gold glitters, so the agent can see it once the agent steps on it.



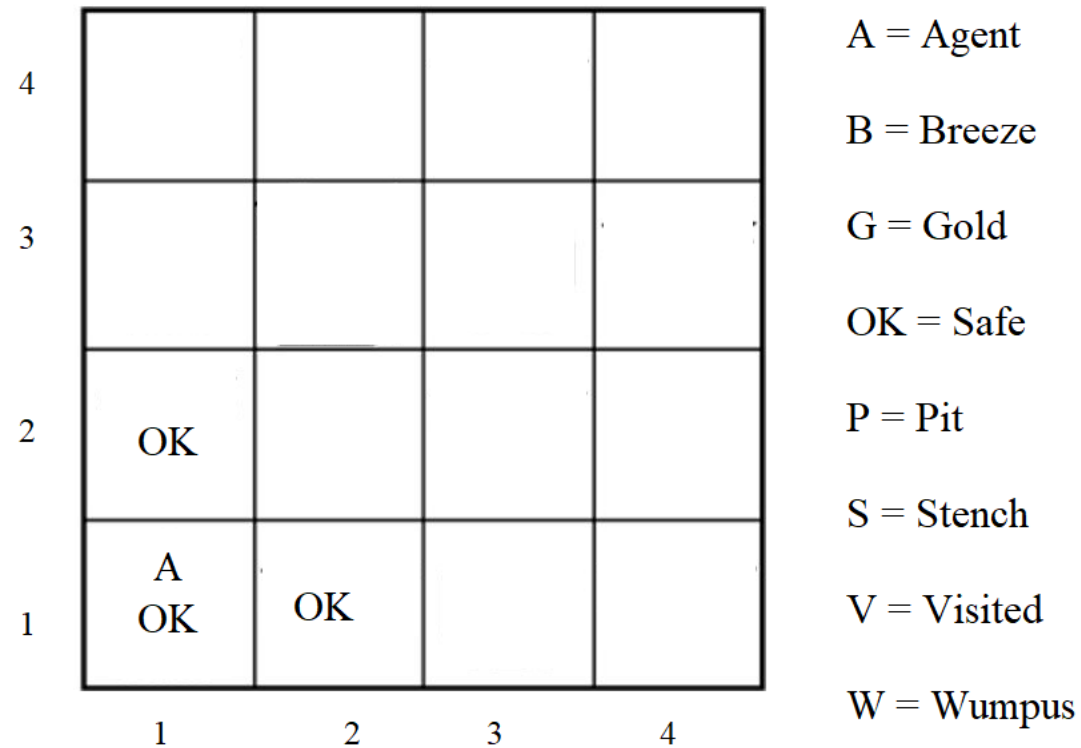
For simplicity, we abstract the game into a 4×4 grid:



Note: The map above is only known by the game designer, but not the agent when she starts.

Step 0: The agent starts at (1,1).

- Knowledge: (1,1) is safe.
- Inference: There is no B nor S at (1,1), so (1,2), (2,1) are safe.



Step 1: The agent moves right to (2,1).

- Knowledge: (2,1) has B.
- Inference:
 - Either (2,2) or (3,1) has a P, or both have.
 - (2,2) and (3,1) do not contain W.

4				
3				
2	OK	P?		
1	OK V	A B OK	P?	
	1	2	3	4

A = Agent

B = Breeze

G = Gold

OK = Safe

P = Pit

S = Stench

V = Visited

W = Wumpus

Step 2: The agent moves to (1,2).

- Knowledge: (1,2) has S.
- Inference:
 - (2,2) has no P, Thus is safe.
 - (1,3) must has W.

4				
3	W			
2	A S OK	OK		
1	OK V	B V OK	P	
	1	2	3	4

A = Agent

B = Breeze

G = Gold

OK = Safe

P = Pit

S = Stench

V = Visited

W = Wumpus

Step 3: The agent moves to (2,2).

- Knowledge: (2,2) has no B nor S.
- Inference: (2,3), (3,2) are both safe.

4				
3	W	OK		
2	V S OK	A OK	OK	
1	OK V	B V OK	P	
	1	2	3	4

A = Agent

B = Breeze

G = Gold

OK = Safe

P = Pit

S = Stench

V = Visited

W = Wumpus

Step 4: The agent moves to (2,3).

- Knowledge: (2,3) has B, S and G

4		P ?		
3	W	A G S B OK	P ?	
2	V S OK	V OK	OK	
1	OK V	B V OK	P	
	1	2	3	4

A = Agent

B = Breeze

G = Gold

OK = Safe

P = Pit

S = Stench

V = Visited

W = Wumpus

In a **knowledge-based model**, the world is represented by **knowledge**, i.e., facts that the agents perceives, infers, and remembers. An **knowledge-based agent** consists of two parts:

- **Offline**: the agent stores a **knowledge base (KB)**. The knowledge base stores previously held facts.

E.g. “A wumpus is stink and all locations around it will have stench”

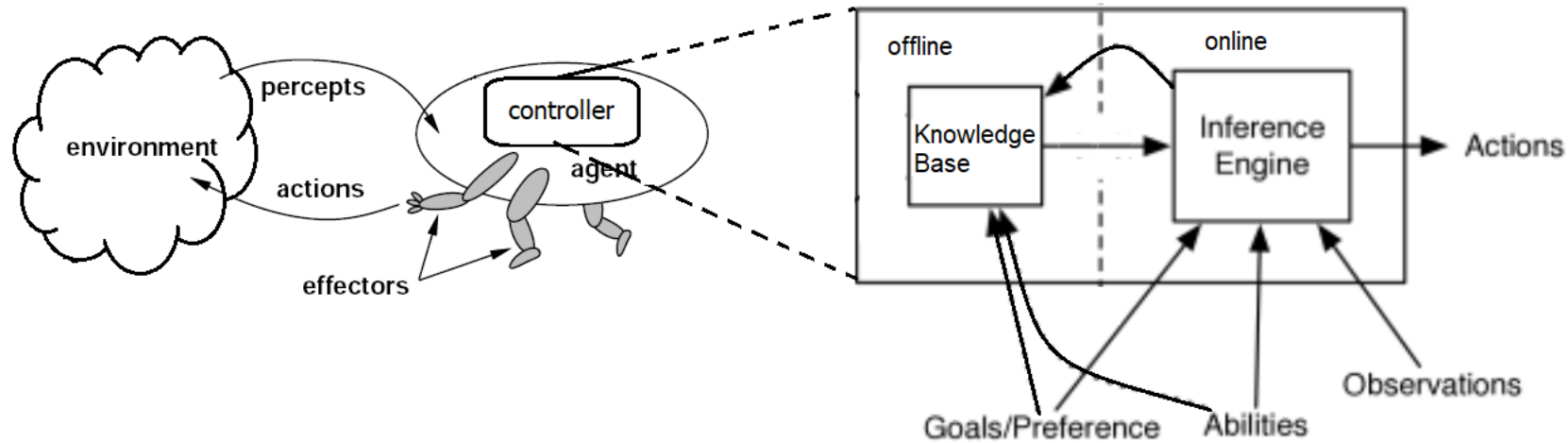
“A pit has air through it so all locations around it will have breeze”

- **Online**: the agent implements an **inference engine (IE)**, i.e., algorithms that generate new facts from existing facts.

E.g. “Local (i, j) has no breeze nor stench. Thus $(i, j + 1)$ and $(i + 1, j)$ are both safe”.

“Given $(1,1)$ has no wumpus, since $(2,1)$ has no stench but $(1,2)$ has stench, $(1,3)$ must has wumpus”.

Knowledge-based agent architecture:



To describe a knowledge-based model, we need to answer the following questions:

- How do we represent knowledge?
- How do we implement an inference engine?

Knowledge Representations

Question. How do we represent knowledge?

- Knowledge as **statements**.

E.g.

- ‘There is breeze here.’
- ‘At least one of these two locations must be a pit.’
- ‘If there is stench, then a wumpus is nearby.’

- Knowledge as **constraints**.

E.g.

- ‘There is breeze here’ restricts the possible worlds to have breeze in the current location.
- ‘At least one of these two locations must be a pit’ restricts the possible worlds to have pit in either one or both of the two locations.
- ‘If there is a stench, then a wumpus is nearby’ restricts the possible worlds to link stench with wumpus.

Knowledge representation methods

There are different approaches to representing knowledge in AI systems:

- **Logical Representation:**

- Use mathematical **logic** to organize and represent knowledge.
- Define a set of symbols, such as **variables, constants, predicates, and logical connectives**
- A set of **inference rules** to perform logical deductions on these symbols.

- **Semantic Networks:**

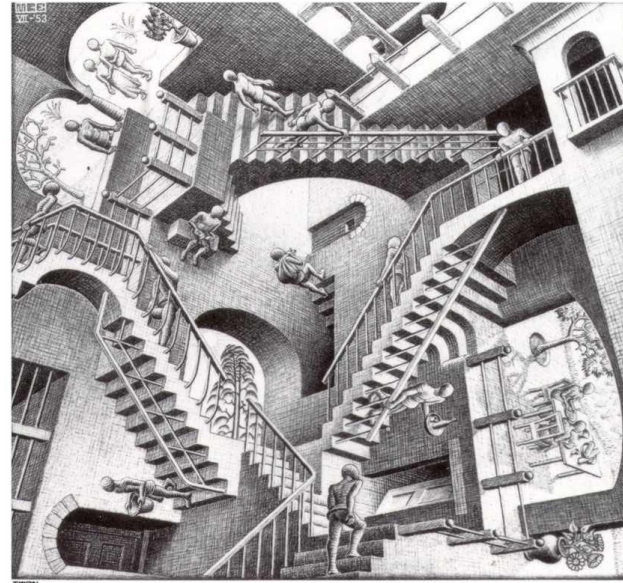
- Knowledge is represented as nodes (representing **concepts** or **objects**) connected by links (indicating **relationships between concepts**).
- Semantic networks are particularly useful for visualizing complex relationships between entities.

- Other approaches: frames and rules, etc.

Logical Representation

We need a logical language framework that

- The framework can help us to express statement.
- The statements can be interpreted as constraints.
- An inference engine must be developed based on this framework that generates new statements based on **logical rules**.



Logical Representation

How to define a logical language?

- Atomic level:
 - Syntax: What are the most elementary entities (i.e., atoms) in this language?
 - Semantic: How to interpret these atomic entities?
- Compound level:
 - Syntax: How to construct statements in this language?
 - Semantic: What is the meaning of a statement in this language?

Propositional Logic

Propositional logic provides a simple formal language framework. In this framework, we may specify:

- **Atomic propositions:** k binary features $X_1, \dots, X_k$ with domain $\{\text{true}, \text{false}\}$ (or $\{0,1\}$).
- **Proposition:** Facts regarding these binary features.

Example. We have the following atomic propositions

- A : “It is raining”
- B : “The sun is shining”
- C : “There is rainbow”

Then $((A \wedge B) \rightarrow C)$ represents ‘If it is raining and the sun is shining, then there is a rainbow’



Propositional Logic: Formal Syntax and Semantics

Atomic Entities

- **Syntax:** **Atomic propositions** (or **atoms**) $\text{Atom} = \{X_1, \dots, X_k\}$
- **Semantics:** An **interpretation** is a function $\pi: \text{Atom} \rightarrow \{\text{true}, \text{false}\}$, i.e., π is a possible world.

Compound Entities

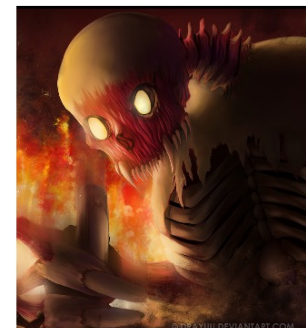
- **Syntax:** A **proposition** can be made up from the atomic propositions in the form
 $\neg A, (A \vee B), (A \wedge B), (A \rightarrow B), (A \leftarrow B), (A \leftrightarrow B)$
where A, B are propositions, $\neg, \wedge, \vee, \rightarrow, \leftarrow, \leftrightarrow$ are **connectives**.
- **Semantics:** An interpretation determines the truth value of all propositions:

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \leftarrow B$	$A \rightarrow B$	$A \leftrightarrow B$
true	true	false	true	true	true	true	true
true	false	false	false	true	true	false	false
false	true	true	false	true	false	true	false
false	false	true	false	false	true	true	true

Example. [Wumpus world] There can be number of binary features: For any $i, j \in \{1,2,3,4\}^2$

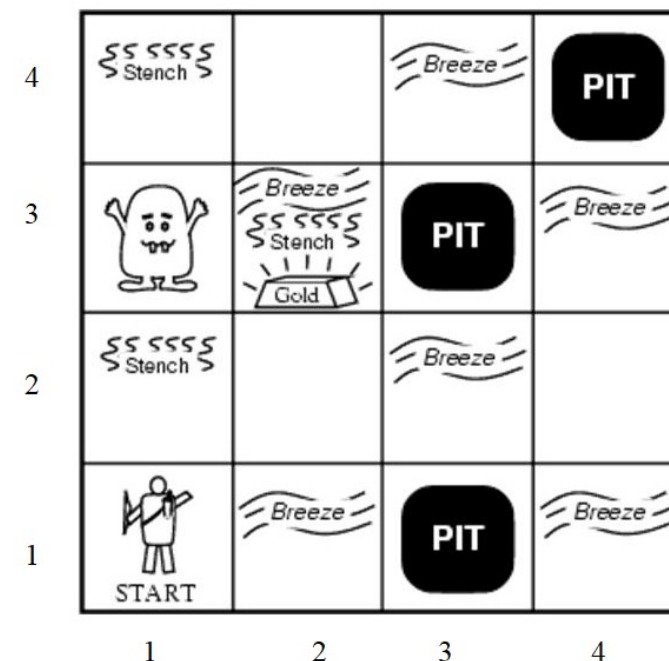
- $A_{i,j}$ is true if and only if the agent locates in (i, j)
- $B_{i,j}$ is true if and only if (i, j) has breeze
- $G_{i,j}$ is true if and only if (i, j) contains gold
- $Ok_{i,j}$ is true if and only if (i, j) is safe
- $P_{i,j}$ is true if and only if (i, j) has a pit
- $V_{i,j}$ is true if and only if (i, j) has been visited by the agent
- $W_{i,j}$ is true if and only if (i, j) has wumpus

These are the atomic propositions.



We can use formal, symbolic expressions for facts about the Wumpus world:

- ‘(1,3) has a pit’: $P_{1,3}$
- ‘(2,2) has no wumpus’: $\neg W_{2,2}$
- ‘Either (2,2) has a pit or (3,1) has a pit’: $P_{2,2} \vee P_{3,1}$
- ‘Since (2,2) has no stench, (1,2) has no wumpus’: $\neg S_{2,2} \rightarrow \neg W_{1,2}$
- ‘(2,4) is safe if and only if there is no pit nor wumpus on (2,4)’: $OK_{2,4} \leftrightarrow (\neg P_{2,4} \wedge \neg W_{2,4})$



We can make inference from facts about the wumpus world:

- $P_1 = \text{'There is no stench and no breeze in (1,1)'}: \neg S_{1,1} \wedge \neg B_{1,1}$
- $P_2 = \text{'There is no stench in (1,1)'}: \neg S_{1,1}$

Note that from P_1 , we can infer P_2 .

- $P_3 = \text{'If (3,1) is a pit, then both (2,1) and (3,2) have breeze'}$
 $P_{3,1} \rightarrow (B_{2,1} \wedge B_{3,2})$
- $P_4 = \text{'If one of (2,1) or (3,2) has no breeze, then (3,1) is not a pit'}$
 $(\neg B_{2,1} \vee \neg B_{3,2}) \rightarrow \neg P_{3,1}$

Note that from P_3 , we can infer P_4 , and vice versa.

Definition

A proposition A **logically implies** B , written $A \Rightarrow B$, if for any interpretation π ,
if $\pi(A) = \text{true}$ then $\pi(B) = \text{true}$

Example.

- $\neg S_{1,1} \wedge \neg B_{1,1} \Rightarrow \neg S_{1,1}$
- $P_{3,1} \rightarrow (B_{2,1} \vee B_{3,2}) \Rightarrow (\neg B_{2,1} \vee \neg B_{3,2}) \rightarrow \neg P_{3,1}$
- $(\neg B_{2,1} \vee \neg B_{3,2}) \rightarrow \neg P_{3,1} \Rightarrow P_{3,1} \rightarrow (B_{2,1} \vee B_{3,2})$
- (**Modus ponens**) For any A, B , $((A \rightarrow B) \wedge A) \Rightarrow B$
- (**Modus tollens**) For any A, B , $((A \rightarrow B) \wedge \neg B) \Rightarrow \neg A$
- (**Syllogism**) For any A, B, C , $((A \rightarrow B) \wedge (B \rightarrow C)) \Rightarrow (A \rightarrow C)$

Two propositions A, B are **logically equivalent**, written $A \Leftrightarrow B$, if $A \Rightarrow B$ and $B \Rightarrow A$

$\neg\neg A$	$\Leftrightarrow$	A	Double negation
$(A \wedge B)$	$\Leftrightarrow$	$(B \wedge A)$	Commutative law
$(A \vee B)$	$\Leftrightarrow$	$(B \vee A)$	
$(A \wedge (B \wedge C))$	$\Leftrightarrow$	$((A \wedge B) \wedge C)$	Associative law
$(A \vee (B \vee C))$	$\Leftrightarrow$	$((A \vee B) \vee C)$	
$(A \wedge (B \vee C))$	$\Leftrightarrow$	$((A \wedge B) \vee (A \wedge C))$	Distributive law
$(A \vee (B \wedge C))$	$\Leftrightarrow$	$((A \vee B) \wedge (A \vee C))$	
$(A \wedge A)$	$\Leftrightarrow$	A	Idempotent law
$(A \vee A)$	$\Leftrightarrow$	A	
$\neg(A \wedge B)$	$\Leftrightarrow$	$(\neg A \vee \neg B)$	De Morgan's law
$\neg(A \vee B)$	$\Leftrightarrow$	$(\neg A \wedge \neg B)$	
$(A \rightarrow B)$	$\Leftrightarrow$	$(\neg A \vee B)$	Implication law
$(A \rightarrow B)$	$\Leftrightarrow$	$\neg(A \wedge \neg B)$	
$(A \rightarrow B)$	$\Leftrightarrow$	$B \leftarrow A$	
$(A \vee (B \wedge \neg B))$	$\Leftrightarrow$	A	Contradiction
A	$\Leftrightarrow$	$(A \wedge (A \vee B))$	Absorption law
A	$\Leftrightarrow$	$(A \vee (A \wedge B))$	
$(A \leftrightarrow B)$	$\Leftrightarrow$	$((A \rightarrow B) \wedge (B \rightarrow A))$	Equivalence law
$(A \leftrightarrow B)$	$\Leftrightarrow$	$((A \wedge B) \vee (\neg A \wedge \neg B))$	

- An interpretation π **satisfies a proposition A** if $\pi(A) = \text{true}$.
- An interpretation π **satisfies a set S of propositions** if π satisfies every $A \in S$.

Definition

Suppose $X_1, \dots, X_k$ are atomic propositions.

- A **literal** is X_i or $\neg X_i$.
- A **clause** is a proposition of the form

$$(h_1 \vee h_2 \vee \dots \vee h_m) \leftarrow (l_1 \wedge l_2 \wedge \dots \wedge l_k)$$
 where $m \geq 1$ and $k \geq 0$, each h_i and each l_j is a literal.
- $(h_1 \vee h_2 \vee \dots \vee h_m)$ is called the **head** of the clause.
 $(l_1 \wedge l_2 \wedge \dots \wedge l_k)$ is called the **body** of the clause.

E.g. $\neg C \leftarrow (\neg A \wedge B)$ is a clause. $C \leftarrow (A \vee B)$ is not.

A clause represents a **constraint**: An interpretation π satisfies

$$(h_1 \vee h_2 \vee \cdots \vee h_m) \leftarrow (l_1 \wedge l_2 \wedge \cdots \wedge l_k)$$

If and only if π satisfies

$$h_1 \vee h_2 \vee \cdots \vee h_m \vee \neg l_1 \vee \neg l_2 \vee \cdots \vee \neg l_k$$

If and only if π does not satisfies

$$\neg h_1 \wedge \neg h_2 \wedge \cdots \wedge \neg h_m \wedge l_1 \wedge l_2 \wedge \cdots \wedge l_k$$

This is equivalent to say that $h_1, \dots, h_m, \neg l_1, \dots, \neg l_k$ should not be false in π together.

Propositional Knowledge Base

- A **propositional knowledge base** is a set of clauses can be regarded as **knowledge** about the atomic propositions.
- For any proposition A , there is a set of clauses S such that any interpretation π satisfies A if and only if π satisfies S .

Definition

A **knowledge base** is a set of clauses.

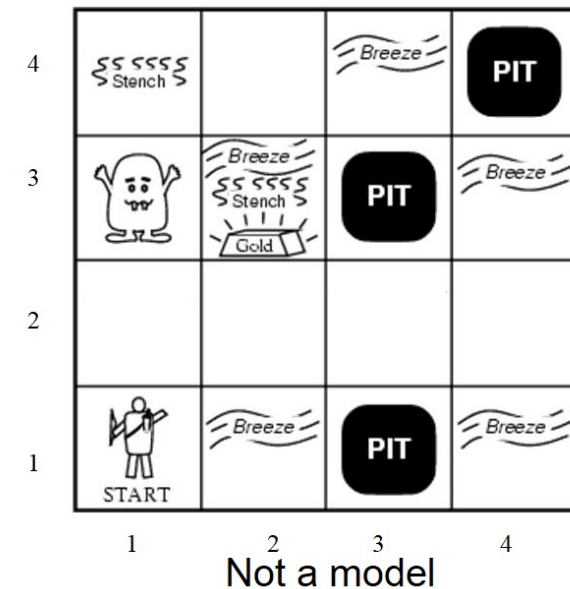
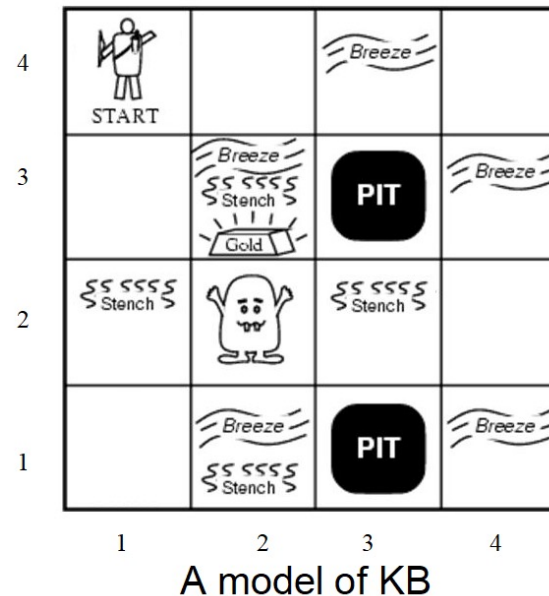
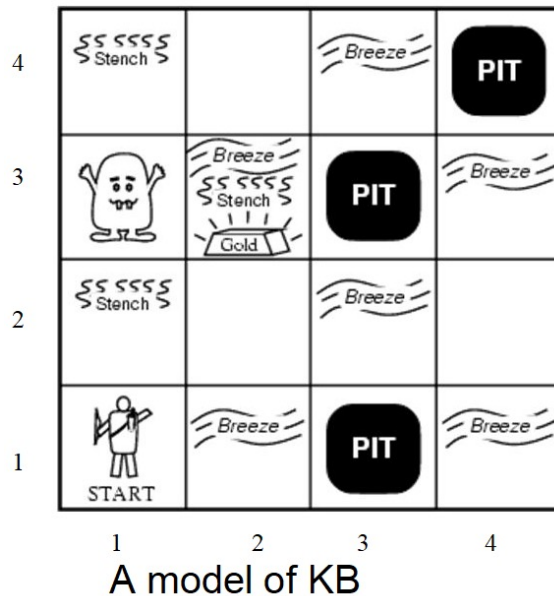
Definition

A **model** of a knowledge base KB is an interpretation that satisfies KB.

Example. [Wumpus world] KB contains

- $S_{i',j'} \leftarrow W_{i,j}$ and $B_{i',j'} \leftarrow P_{i,j}$ where $(i,j) \in \{1, \dots, 4\}^2$ and $|i' - i| + |j' - j| = 1$.
- $(W_{i-1,j} \vee W_{i,j-1} \vee W_{i+1,j} \vee W_{i,j+1}) \leftarrow S_{i,j}$ and $(P_{i-1,j} \vee P_{i,j-1} \vee P_{i+1,j} \vee P_{i,j+1}) \leftarrow B_{i,j}$ for $i,j \in \{2,3\}^2$.
- $(W_{2,1} \vee W_{1,2}) \leftarrow S_{1,1}$, $(P_{2,1} \vee P_{1,2}) \leftarrow B_{1,1}$, ... which describe if a location has wumpus or pit, then the surround locations have stench or breeze, respectively.

The following represents three interpretations:



Properties of Logical Language

Logic as a knowledge representation language:

- It is **declarative**: Separation between knowledge and control
E.g., The language describes what an agent senses in the wumpus world, not how to achieve its goals.
- It is **expressive**: Ability to handle unknown information or partially-known information.
E.g., The language can state “there is a pit in [2,2] or [3,1]”, or “if the wumpus is in [1,1], then it is not in [2,2]”.
- It is **compositional**: The language can compose sentences into longer sentences
E.g., The meaning of $B_{1,1} \wedge P_{1,2}$ is derived from meaning of $B_{1,1}$ and of $P_{1,2}$

Limitations of propositional logic

- **Verbosity**: Stating simple facts requires listing many propositions

E.g., “A breeze is sensed if and only if an adjacent location contains a pit”

$$B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1}), B_{1,2} \leftrightarrow (P_{1,3} \vee P_{2,2} \vee P_{1,1}), B_{2,1} \leftrightarrow (P_{2,2} \vee P_{3,1} \vee P_{1,1}) \dots$$

- **Low expressiveness**: Many facts cannot be expressed by propositional logic

E.g.,

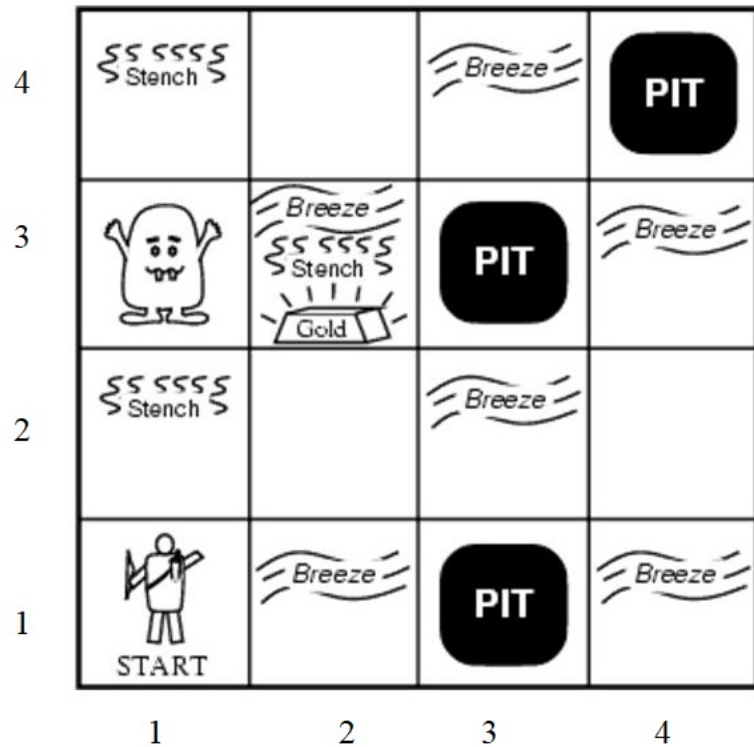
- **Universal sentence**: “Every mammal is breast-feeding”
- **Existential sentence**: “Some birds does not fly”
- **Reasoning by instantiation**: “All mammals are breast-feeding. A dolphin is a mammal. Therefore dolphins are breast-feeding”

First-Order Logic (FOL)

Whereas propositional logic assumes the world contains **facts**, first-order logic (like natural language) assumes the world contains

- **Objects:** people, houses, numbers, colors, baseball games, wars, ...
- **Relations:**
 - Unary relations: being red, being smart, being a prime number
 - Binary relations: bigger than, older than, divides
 - n -ary relations ($n \geq 3$): mother, father and child relation
- **Functions:** father of, best friend, one more than, plus, ...

Example. How would you describe the state of the wumpus world below?



- **Domain:** A set of objects

E.g., $\{(1,1), (1,2), \dots, (4,3), (4,4)\}$ representing the 4x4 environment

- **Relations:**

- **Unary relation:** Breeze = $\{(1,2), (1,4), (2,3), (3,2), (3,4), (4,3)\}$

Pit = $\{(1,3), (3,3), (4,4)\}$, Stench = $\{(2,1), (4,1), (3,2)\}$

WumpusAt = $\{(3,1)\}$, AgentAt = $\{(1,1)\}$, Glitter = $\{(3,2)\}$

- **Binary relation:** Adjacent = $\{((1,1), (1,2)), ((1,1), (2,1)), \dots\}$

- **Functions:**

- $left((1,2)) = (1,1)$

- $right((2,3)) = (2,4)$

First-Order Logic: Formal Semantics

FOL Semantics

A **first-order interpretation** is a tuple

$$(D, R_1, R_2, \dots, R_k, f_1, \dots, f_l)$$

where

- D is a set of elements, which is called domain.
- Each R_i is a relation (or predicate), defined on D
- Each f_i is a function, defined on D

A **first-order interpretation** describes a **state** of the current world

First-Order Logic: Formal Syntax

FOL Alphabet

The **alphabet** of a first-order language contains the following symbols:

- **Constant:** $c_1, c_2, c_3 \dots$
- **Variables:** $v_1, v_2, v_3, \dots$
- **Connectives:** $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
- **Quantifiers:** $\forall, \exists$
- **Equality:** $=$
- **Parentheses:** $(,)$
- **Relational (or predicate) symbols:** $R_1, R_2 \dots, R_k$
- **Functional symbols:** $f_1, f_2, \dots, f_l$

Definition

The set $S = \{R_1, R_2, \dots, R_k, f_1, f_2, \dots, f_l\}$ of relational symbols and functional symbols in the alphabet of a first order language is called its **signature**.

E.g., Signature of the Wumpus world could be

AgentAt, WumpusAt, Pit, Breeze, Stench, Glitters, Adjacent, left, right, up, down

Definition

Let S be a signature. The set of terms T are defined as follows:

- Every **variable** is a term in T
- Every **constant** is a term in T .
- If $t_0, t_1, \dots, t_{r-1}$ are terms and f is a function symbol in S , then $f(t_0, t_1, \dots, t_{r-1})$ is a term in T .

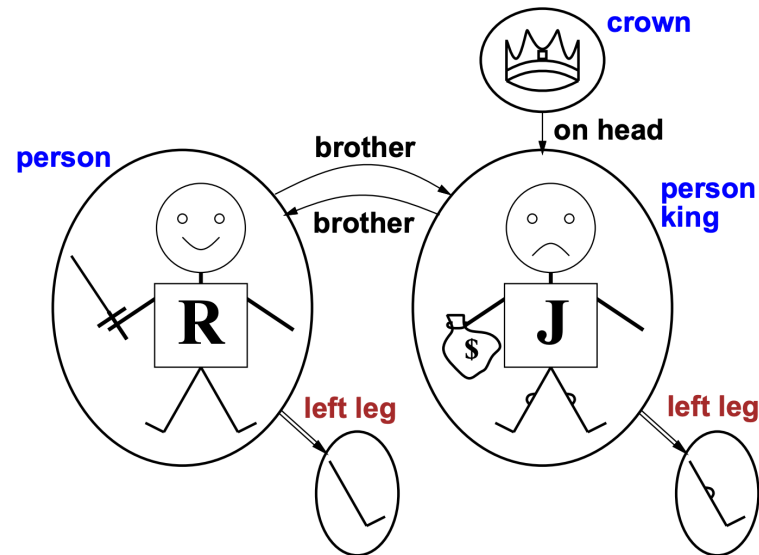
A **ground term** is a term that does not contain any variable symbol.

E.g., $(1,0)$, $down(right(down(3,2)))$ are ground terms, $left(up(x, y))$ is not a ground term

First-Order Logic: Formal Syntax

Definition

- An **atomic sentence** is formed by a predicate symbol, followed by a parenthesized list of terms, or the equality between two terms.
- A **complex sentence** is formed by multiple atomic sentences connected by the logical connectives $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$.



Example of atomic sentences:

Brother(Richard, John)

Married(Father(Richard), Mother(John))

Father(John) = Henry

Example of complex sentences:

$\neg \text{Brother}(\text{LeftLeg}(\text{Richard}), \text{John})$

$\text{Brother}(\text{Richard}, \text{John}) \wedge \text{Brother}(\text{John}, \text{Richard})$

$\text{King}(\text{Richard}) \vee \text{King}(\text{John})$

Truth in FOL

Sentences are true with respect to a **model** and an **interpretation**

Model contains objects (**domain elements**) and relations among them

Interpretation specifies referents for

constant symbols	→	objects
predicate symbols	→	relations
function symbols	→	functional relations

An atomic sentence $predicate(term_1, \dots, term_n)$ is true
iff the **objects** referred to by $term_1, \dots, term_n$
are in the **relation** referred to by $predicate$

Truth example

Consider the interpretation in which

Richard → **Richard the Lionheart**

John → **the evil King John**

Brother → **the brotherhood relation**

Under this interpretation,

Brother(Richard, John) is true, just in case **Richard the Lionheart** and the **evil King John** are in the **brotherhood** relation in the model

Universal quantification

$\forall \langle \text{variables} \rangle \langle \text{sentence} \rangle$

Everyone at HIT is smart:

$\forall x \text{ At}(x, \text{HIT}) \rightarrow \text{Smart}(x)$

$\forall x P$ is true in a model m iff P is true with x being **each** possible object in the model

Roughly speaking, equivalent to the **conjunction** of **instantiations** of P

$$\begin{aligned} & \text{At}(\text{KingJohn}, \text{HIT}) \rightarrow \text{Smart}(\text{KingJohn}) \\ \wedge & \text{At}(\text{Richard}, \text{HIT}) \rightarrow \text{Smart}(\text{Richard}) \\ \wedge & \text{At}(\text{HIT}, \text{HIT}) \rightarrow \text{Smart}(\text{HIT}) \\ \wedge & \dots \end{aligned}$$

A common mistake to avoid:

Typically, $\rightarrow$ is the main connective with $\forall$

Common mistake: using $\wedge$ as the main connective with $\forall$:

$$\forall x \text{ } At(x, HIT) \wedge Smart(x)$$

means “Everyone is at HIT and everyone is smart”

Existential quantification

$\exists \langle \text{variables} \rangle \langle \text{sentence} \rangle$

Someone at HIT is smart:

$\exists x \text{ At}(x, \text{HIT}) \wedge \text{Smart}(x)$

$\exists x \text{ } P$ is **true** in a model m iff P is **true** with x being **some** possible object in the model

Roughly speaking, equivalent to the **disjunction** of **instantiations** of P

$\begin{aligned} & \text{At}(\text{KingJohn}, \text{HIT}) \wedge \text{Smart}(\text{KingJohn}) \\ \vee & \text{At}(\text{Richard}, \text{HIT}) \wedge \text{Smart}(\text{Richard}) \\ \vee & \dots \end{aligned}$

Another common mistake to avoid:

Typically, $\wedge$ is the main connective with $\exists$

Common mistake: using $\rightarrow$ as the main connective with $\exists$:

$$\exists x \text{ At}(x, \text{HIT}) \rightarrow \text{Smart}(x)$$

is true if there is anyone who is not at HIT!

Properties of quantifiers

Commutativity

$\forall x \forall y$ is the same as $\forall y \forall x$ (why??)

$\exists x \exists y$ is the same as $\exists y \exists x$ (why??)

$\exists x \forall y$ is **not** the same as $\forall y \exists x$

$\exists x \forall y \text{ Loves}(x,y)$

“There is a person who loves everyone in the world”

$\forall y \exists x \text{ Loves}(x,y)$

“Everyone in the world is loved by at least one person”

Quantifier duality: each can be expressed using the other

$\forall x \text{ Likes}(x, \text{IceCream})$ is the same as $\neg \exists x \neg \text{Likes}(x, \text{IceCream})$

$\exists x \text{ Likes}(x, \text{Broccoli})$ is the same as $\neg \forall x \neg \text{Likes}(x, \text{Broccoli})$

Fun with Sentences

Brothers are siblings

Fun with Sentences

Brothers are siblings

$$\forall x,y \textit{ Brother}(x,y) \rightarrow \textit{Sibling}(x,y).$$

“Sibling” is symmetric

Fun with Sentences

Brothers are siblings

$$\forall x,y \text{ Brother}(x,y) \rightarrow \text{Sibling}(x,y)$$

“Sibling” is symmetric

$$\forall x,y \text{ Sibling}(x,y) \leftrightarrow \text{Sibling}(y,x)$$

One’s mother is one’s female parent

Fun with Sentences

Brothers are siblings

$$\forall x,y \text{ Brother}(x,y) \rightarrow \text{Sibling}(x,y)$$

“Sibling” is symmetric

$$\forall x,y \text{ Sibling}(x,y) \leftrightarrow \text{Sibling}(y,x)$$

One’s mother is one’s female parent

$$\forall x,y \text{ Mother}(x,y) \leftrightarrow (\text{Female}(x) \wedge \text{Parent}(x,y))$$

A first cousin is a child of a parent’s sibling

Fun with Sentences

Brothers are siblings

$$\forall x,y \text{ Brother}(x,y) \rightarrow \text{Sibling}(x,y)$$

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$$\forall x,y \text{ Sibling}(x,y) \leftrightarrow \text{Sibling}(y,x)$$

One’s mother is one’s female parent

$$\forall x,y \text{ Mother}(x,y) \leftrightarrow (\text{Female}(x) \wedge \text{Parent}(x,y))$$

A first cousin is a child of a parent’s sibling

$$\forall x,y \text{ FirstCousin}(x,y) \leftrightarrow \exists p,ps \text{ Parent}(p,x) \wedge \text{Sibling}(ps,p) \wedge \text{Parent}(ps,y)$$

Equality

$term_1 = term_2$ is true under a given interpretation
if and only if $term_1$ and $term_2$ refer to the same object

E.g., $1 = 2$ and $\forall x \times(Sqrt(x), Sqrt(x)) = x$ are satisfiable
 $2 = 2$ is valid

E.g., definition of full *Sibling* in terms of *Parent*:

$$\forall x, y \text{ Sibling}(x, y) \leftrightarrow [\neg(x = y) \wedge \exists m, f \neg (m = f) \wedge \text{Parent}(m, x) \wedge \text{Parent}(f, x) \wedge \text{Parent}(m, y) \wedge \text{Parent}(f, y)]$$

Knowledge base for the wumpus world

Perception:

$$\forall b, g, t \text{ Percept}([Smell, b, g], t) \rightarrow Smelt(t)$$

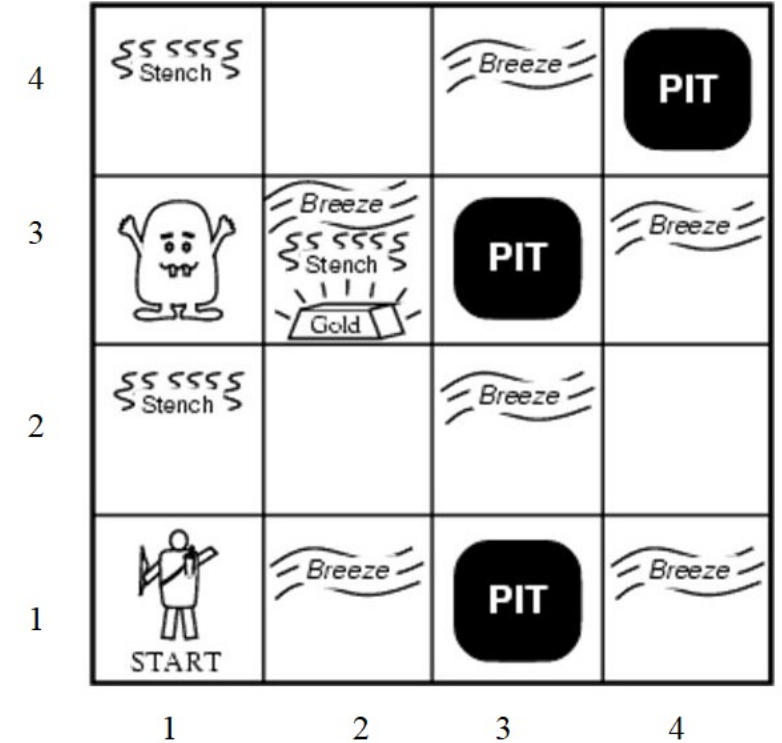
$$\forall s, b, t \text{ Percept}([s, b, Glitter], t) \rightarrow AtGold(t)$$

Reflex:

$$\forall t \text{ AtGold}(t) \rightarrow \text{Action}(\text{Grab}, t)$$

Reflex with internal state: **do we have the gold already?**

$$\forall t \text{ AtGold}(t) \wedge \neg \text{Holding}(\text{Gold}, t) \rightarrow \text{Action}(\text{Grab}, t)$$



Describing the environment

Adjacency:

$$\forall x,y,a,b \text{ Adjacent}([x,y],[a,b]) \leftrightarrow [a,b] \in \{[x+1,y], [x-1,y],[x,y+1],[x,y-1]\}$$

Properties of squares:

$$\forall s,t \text{ AgentAt}(s,t) \wedge \text{Breeze}(t) \rightarrow \text{Breezy}(s)$$

$$\forall s,t \text{ AgentAt}(s,t) \wedge \text{Smelt}(t) \rightarrow \text{Smelly}(s)$$

Squares are breezy near a pit:

- **Diagnostic** rule ----- infer cause from effect

$$\forall s \text{ Breezy}(s) \rightarrow \exists r \text{ Adjacent}(r,s) \wedge \text{Pit}(r)$$

- **Causal** rule ----- infer effect from cause

$$\forall x,y \text{ Pit}(x) \wedge \text{Adjacent}(x,y) \rightarrow \text{Breezy}(y)$$

Neither of these is complete.

e.g., the causal rule doesn't say whether squares far away from pits can be breezy;
the diagnostic rule doesn't say that all pits cause a breeze (just that all breezes are caused by adjacent pits)

Definition for the **Breezy** predicate:

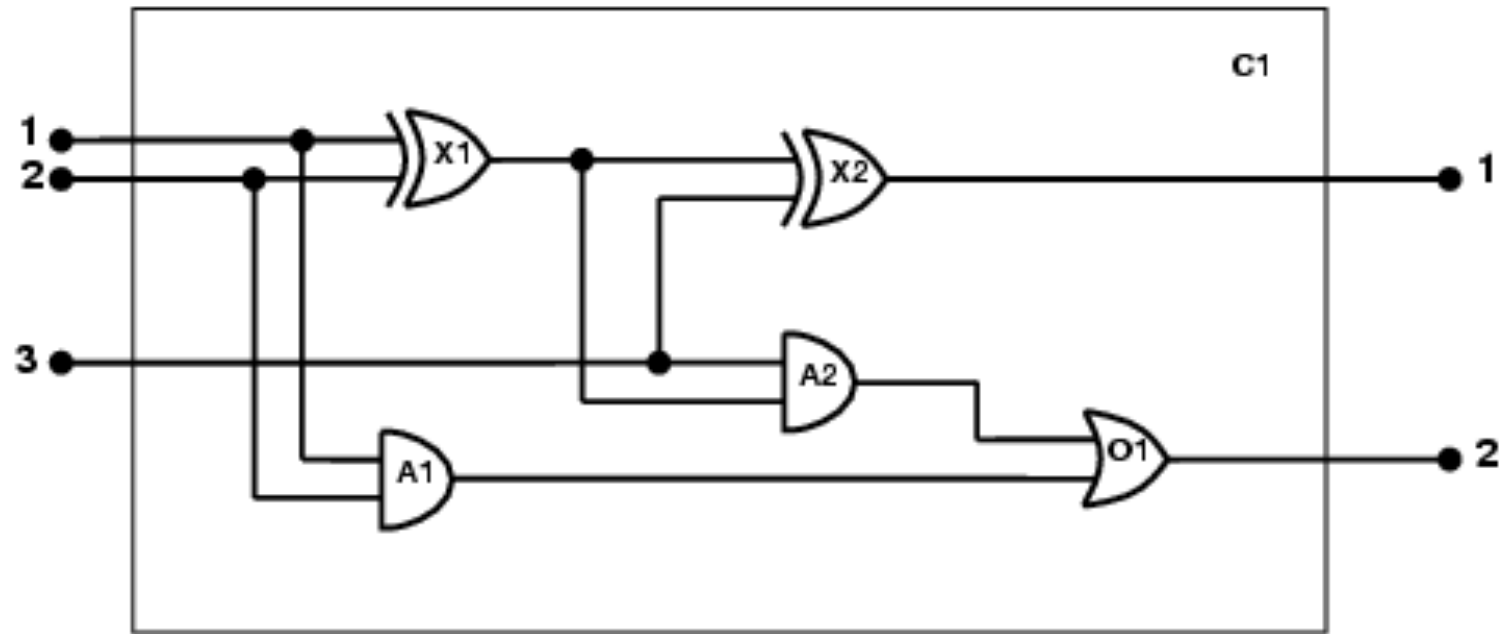
$$\forall y \text{ Breezy}(y) \leftrightarrow [\exists x \text{ Pit}(x) \wedge \text{Adjacent}(x,y)]$$

Knowledge engineering in FOL

1. Identify the task
2. Assemble the relevant knowledge
3. Decide on a vocabulary of predicates, functions, and constants
4. Encode general knowledge about the domain
5. Encode a description of the specific problem instance
6. Pose queries to the inference procedure and get answers
7. Debug the knowledge base

The electronic circuits domain

One-bit full adder



The electronic circuits domain

- 1) Identify the task
 - Does the circuit actually add properly? (circuit verification)
- 2) Assemble the relevant knowledge
 - Composed of wires and gates; Types of gates (*AND*, *OR*, *XOR*, *NOT*)
 - Irrelevant: size, shape, color, cost of gates
- 3) Decide on a vocabulary
- 4) Alternatives:
 - $Type(X_1) = XOR$
 - $Type(X_1, XOR)$
 - $XOR(X_1)$

The electronic circuits domain

5) Encode general knowledge of the domain

- $\forall t_1, t_2 \text{ Connected}(t_1, t_2) \rightarrow \text{Signal}(t_1) = \text{Signal}(t_2)$
- $\forall t \text{ Signal}(t) = 1 \vee \text{Signal}(t) = 0$
- $1 \neq 0$
- $\forall t_1, t_2 \text{ Connected}(t_1, t_2) \rightarrow \text{Connected}(t_2, t_1)$
- $\forall g \text{ Type}(g) = \text{OR} \rightarrow \text{Signal}(\text{Out}(1, g)) = 1 \leftrightarrow \exists n \text{ Signal}(\text{In}(n, g)) = 1$
- $\forall g \text{ Type}(g) = \text{AND} \rightarrow \text{Signal}(\text{Out}(1, g)) = 0 \leftrightarrow \exists n \text{ Signal}(\text{In}(n, g)) = 0$
- $\forall g \text{ Type}(g) = \text{XOR} \rightarrow \text{Signal}(\text{Out}(1, g)) = 1 \leftrightarrow \text{Signal}(\text{In}(1, g)) \neq \text{Signal}(\text{In}(2, g))$
- $\forall g \text{ Type}(g) = \text{NOT} \rightarrow \text{Signal}(\text{Out}(1, g)) \neq \text{Signal}(\text{In}(1, g))$

The electronic circuits domain

6) Encode the specific problem instance

- $Type(X_1) = XOR$ $Type(X_2) = XOR$
- $Type(A_1) = AND$ $Type(A_2) = AND$
- $Type(O_1) = OR$

- $Connected(Out(1,X_1),In(1,X_2))$ $Connected(In(1,C_1),In(1,X_1))$
- $Connected(Out(1,X_1),In(2,A_2))$ $Connected(In(1,C_1),In(1,A_1))$
- $Connected(Out(1,A_2),In(1,O_1))$ $Connected(In(2,C_1),In(2,X_1))$
- $Connected(Out(1,A_1),In(2,O_1))$ $Connected(In(2,C_1),In(2,A_1))$
- $Connected(Out(1,X_2),Out(1,C_1))$ $Connected(In(3,C_1),In(2,X_2))$
- $Connected(Out(1,O_1),Out(2,C_1))$ $Connected(In(3,C_1),In(1,A_2))$

The electronic circuits domain

7) Pose queries to the inference procedure

What are the possible sets of values of all the terminals for the adder circuit?

$$\exists i_1, i_2, i_3, o_1, o_2 \text{ Signal(In}(1, C_1)) = i_1 \wedge \text{Signal(In}(2, C_1)) = i_2 \wedge \text{Signal(In}(3, C_1)) = i_3 \wedge \\ \text{Signal(Out}(1, C_1)) = o_1 \wedge \text{Signal(Out}(2, C_1)) = o_2$$

8) Debug the knowledge base

May have omitted assertions like $1 \neq 0$

Semantic Networks

- Logical representation of knowledge is precise, but it is not intuitive to human
- Semantic networks provide a more visually intuitive representation
 - **Semantic network**: A labelled directed graph used to represent knowledge through concepts and their semantic relations.
 - **Nodes**: concepts, things, objects, behaviors, states, etc.
 - **Directed arcs**: connections or relationships between the nodes.

Semantic Networks

- A semantic unit in semantic networks is often represented by a triplet
(Node1, arc, Node2)

Example: “John is a friend of Tom”

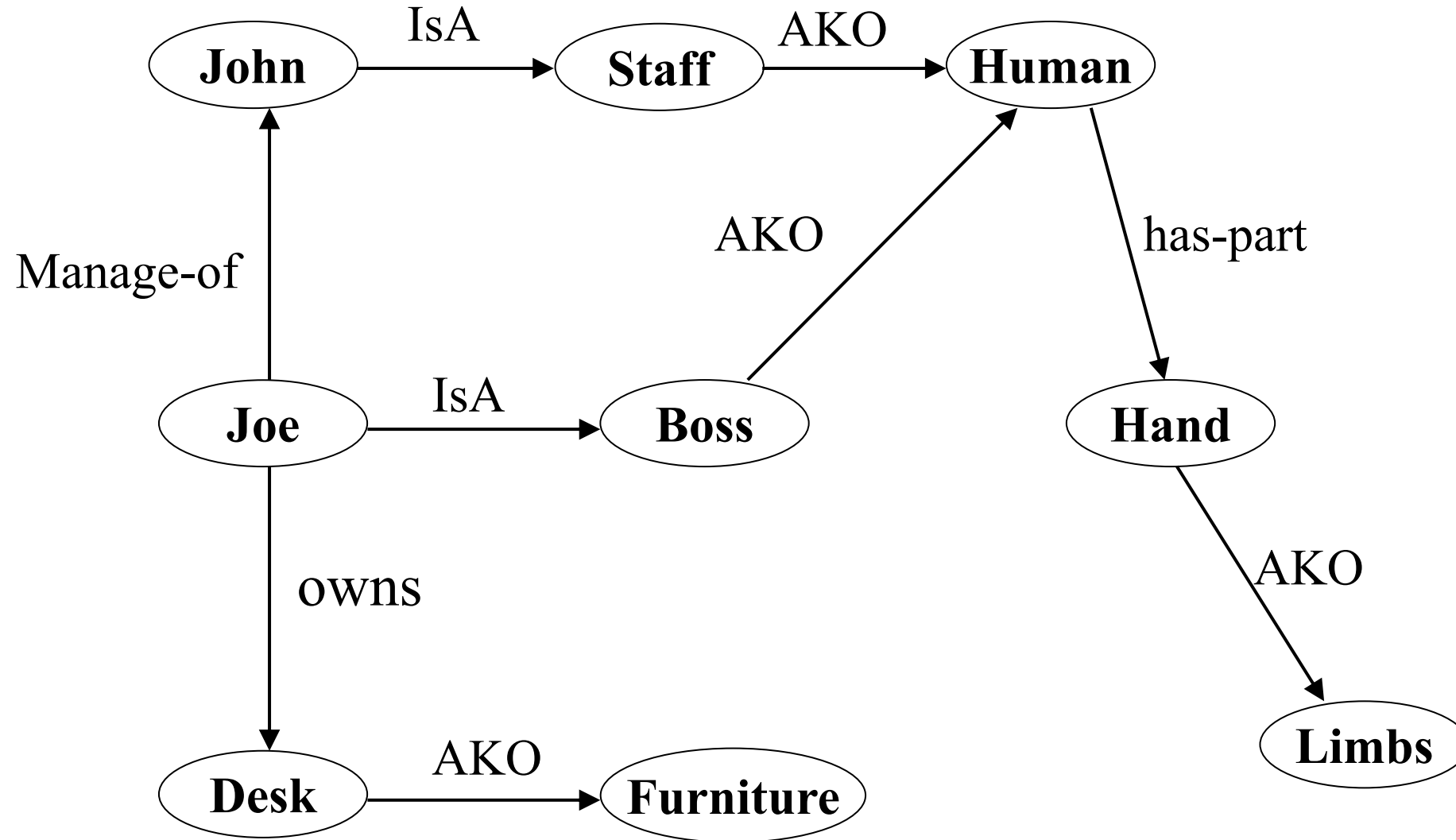
Semantic Network Representation:



FOL Representation:

Friend(John, Tom)

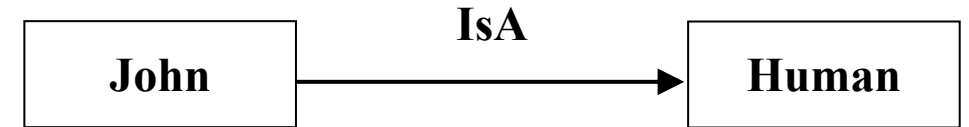
Example:



Basic Relations in Semantic Networks

(1) IsA relation

- Concrete v.s. Abstract
- Indicating that one thing is an instance of another.



(2) AKO (A-Kind-Of) relation

- Subclass v.s. Superclass
- Indicating that one thing is a type of another.



(3) A-Member-of relation

- Individual v.s. Collective
- Indicating that one thing is a member of another.

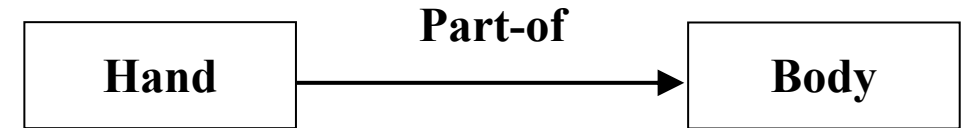


A key property of these relations is the **inheritance of attributes**, where nodes at the concrete level can inherit all the attributes of nodes at all abstract levels.

Basic Relations in Semantic Networks

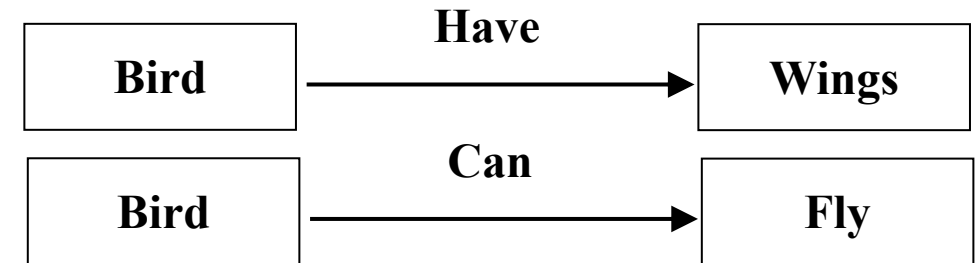
(4) Part-of relation

- Part v.s. the whole
- Indicating that one thing is a part of another.
- This relation does not indicate the **inheritance of attributes**. E.g., Hands don't have all attributes of body



(5) Attribute relation

- Have: Indicating that one node has another.
- Can: Indicating that one node can do another



(6) Temporal relation (Before, After)

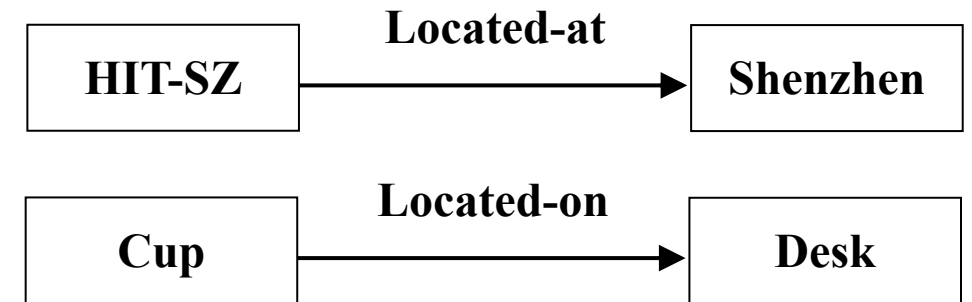
- Indicating that one thing happens before/after another



Basic Relations in Semantic Networks

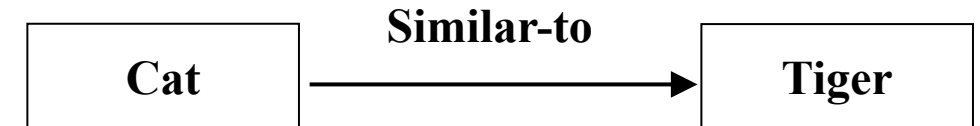
(7) Location relation

- Located-on
- Located-under
- Located-at
- Located-inside
- Located-outside



(8) Proximity relation

- Indicating that one node is similar to another.
- Similar-to, Near-to



Representation of facts and concepts

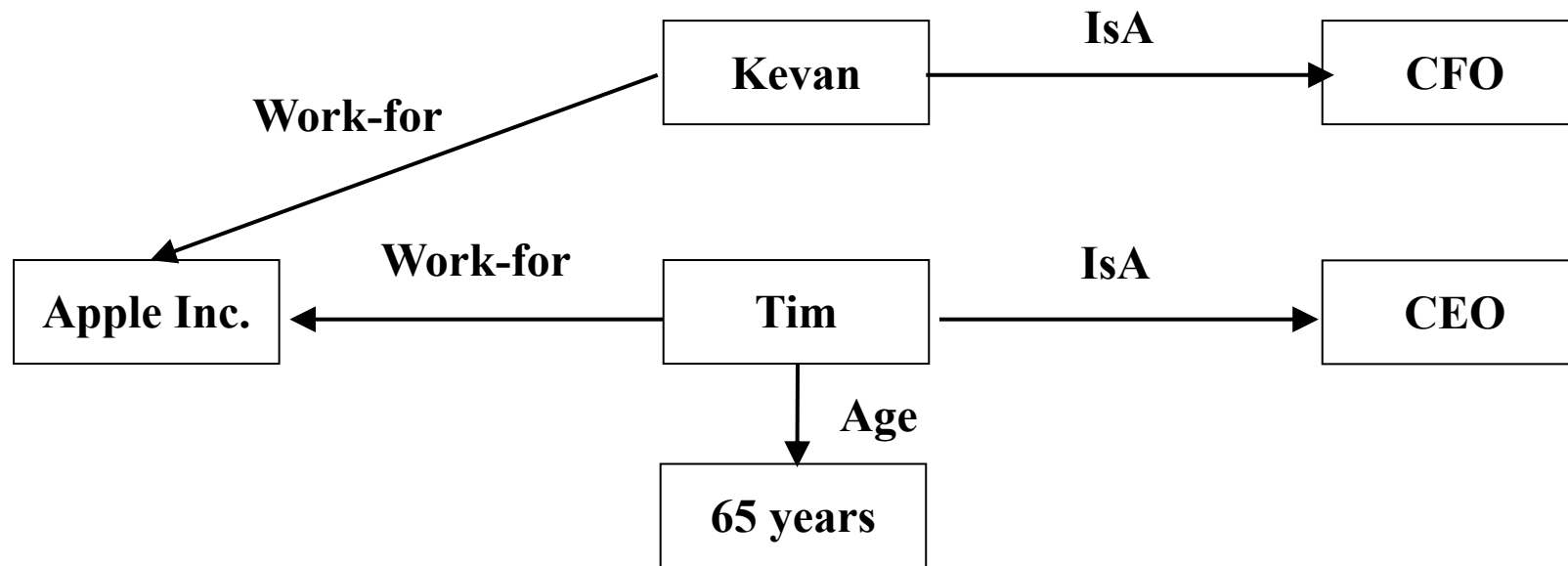
- **Unary relation**: refers to a relationship that can be represented by a unary predicate $P(x)$. The predicate P describes the properties, attributes, etc., of an entity.
 - E.g., HaveWings(Bird) can be represented by the semantic unit (Bird, Have, Wings)
 - Note: Semantic uses binary relation to represent unary relation.
- **Binary relation**: refers to a relationship that can be represented by a binary predicate $P(x,y)$. The predicate P describes the relation between the two entities x,y
 - E.g., FriendOf(John, Tom) can be represented by the semantic unit (John, Friend-of, Tom)

Representation of facts and concepts

Example:

Tim is the CEO of Apple Inc., he is 65 years old.

Kevan is the CFO of Apple Inc.

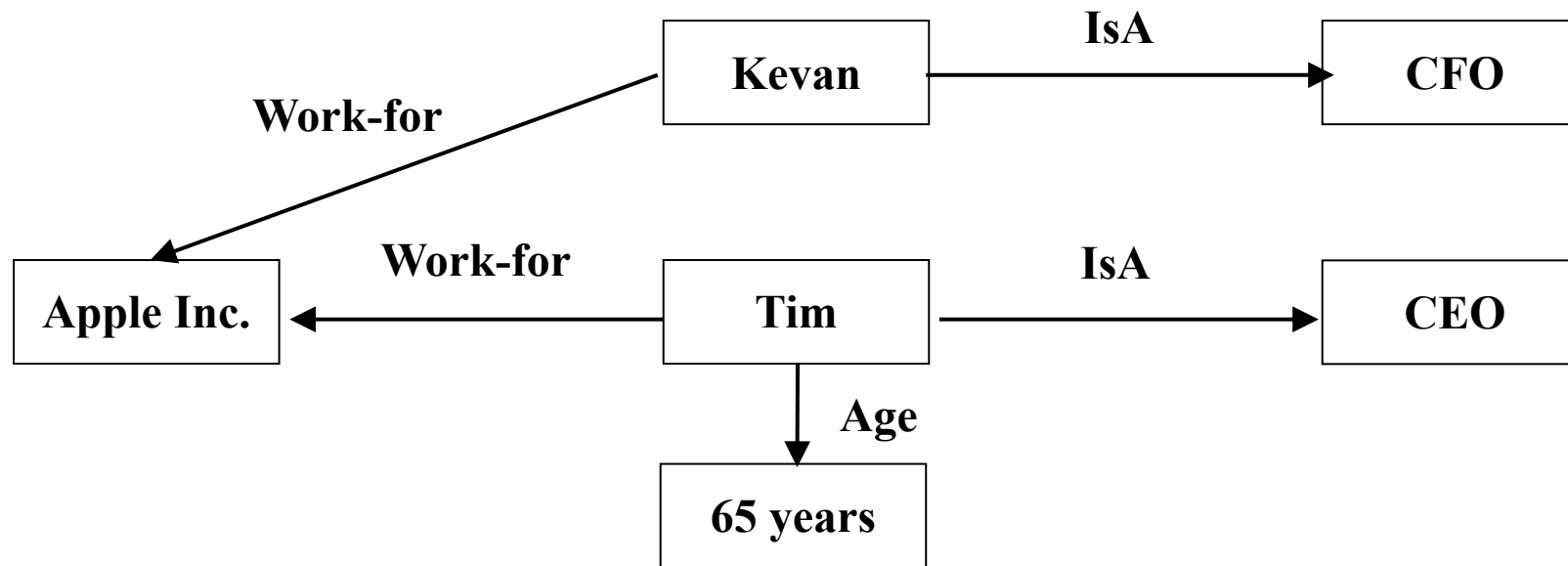


Representation of facts and concepts

Example:

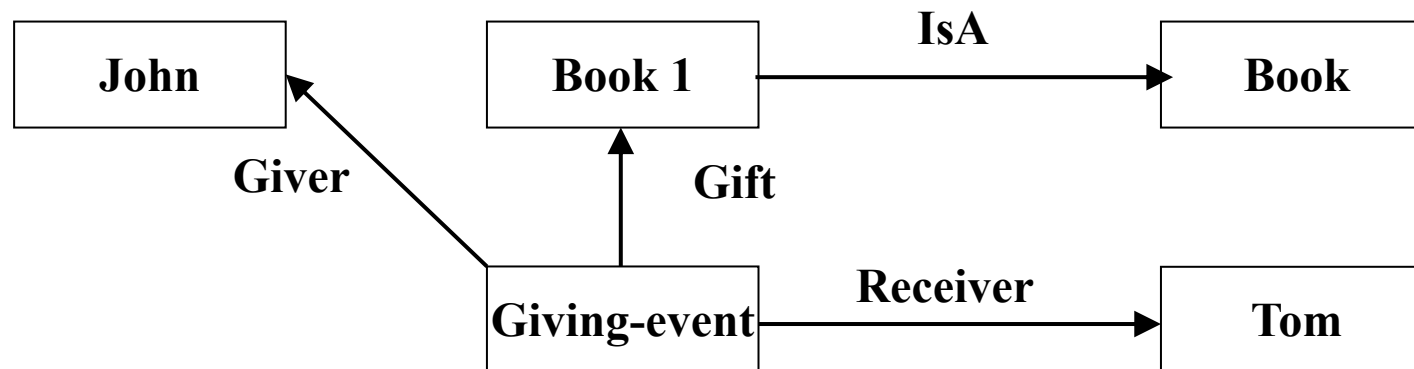
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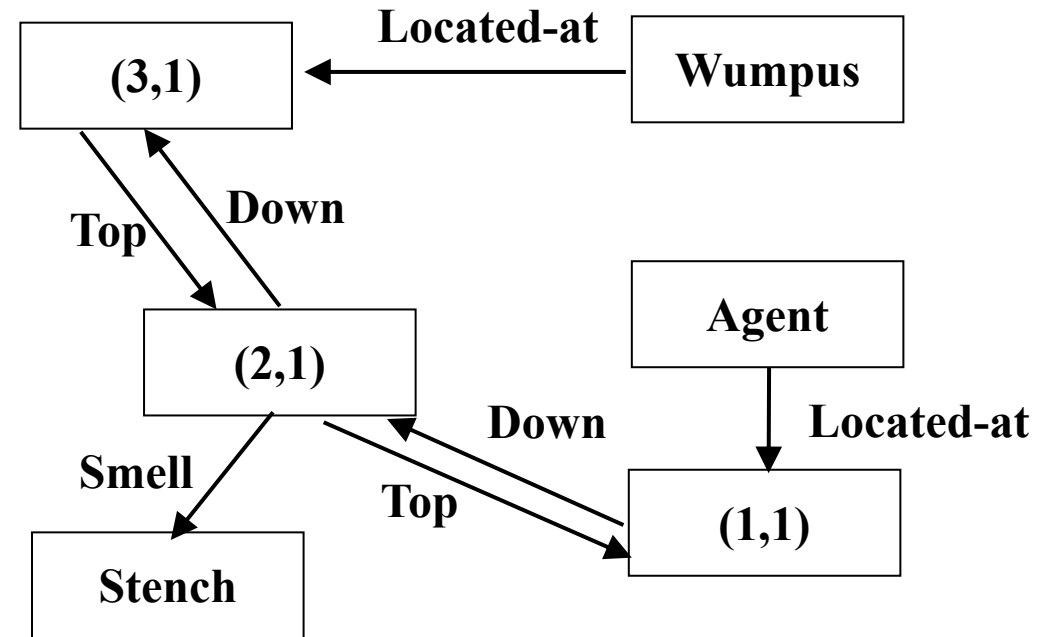
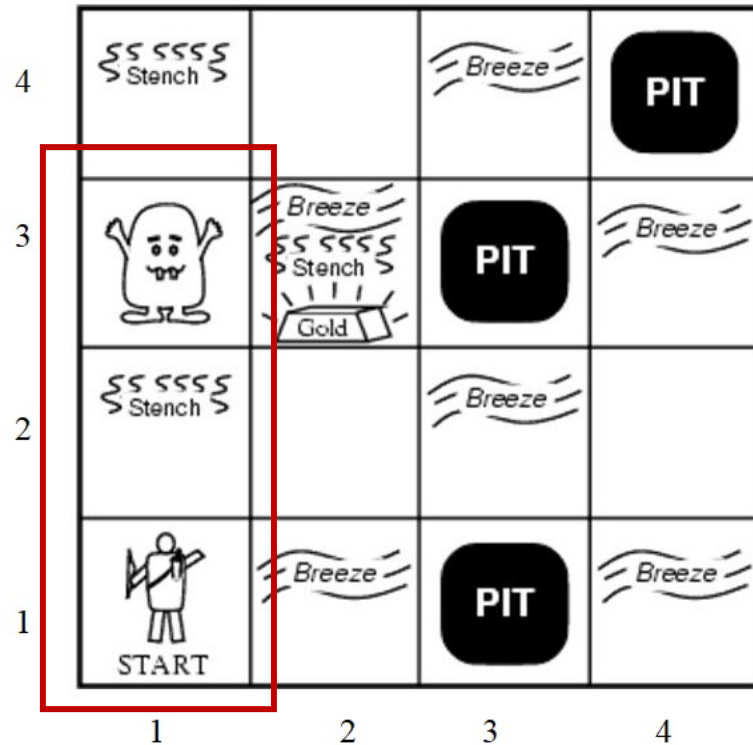


Representation of facts and concepts

- **Multiary relation**: refers to a relationship that can be represented by a multiary predicate $P(x_1, x_2, \dots, x_k)$. The predicate P describes the relation among these entities.
 - E.g., John gives Book1 to Tom, i.e., Give(John, Book1, Tom)
- When representing a multiary relationship using a semantic network, it can be **converted into a combination of one or more binary relationships** to express this multiary relationship.
 - E.g., The “Give” predicate can’t be represented by the semantic unit
 - **Solution**: Create a node “Giving-event” to describe the event



Example. How would you describe the state of the wumpus world below using semantic network?



Part of the semantic network

Summary

- Definition of Knowledge
- Intelligent Agents: Goal-based agent and utility-based agent
- Knowledge in Intelligent Agents
- Knowledge representation methods:

Logical Representation

- Propositional Logic, First-Order Logic
- **Advantages:**
 - Rigor and precise
 - Reasoning based on logical rules
 - Composition of logical expression allows the description of complex knowledge
- **Limitations:**
 - Can't handle uncertainty
 - Not intuitive for human

Semantic Networks

- Use nodes to denote concepts and directed arcs to denote relations
- **Advantages:**
 - Intuitive to human
 - Can handle incomplete and uncertain information
 - Easy to integrate to other systems
- **Limitations:**
 - Semantic is not well-defined
 - No standard definition